SOLUTION

$a = 2t - 6$

$dv = a\,dt$

$$\int_0^v dv = \int_0^t (2t - 6)\,dt$$

$v = t^2 - 6t$

$ds = v\,dt$

$$\int_0^s ds = \int_0^t (t^2 - 6t)\,dt$$

$$s = \frac{t^3}{3} - 3t^2$$

When $t = 6\,s$,

$v = 0$ **Ans.**

When $t = 11\,s$,

$s = 80.7\text{ m}$ **Ans.**

Ans:

$s = 80.7\text{ m}$

SOLUTION

$$s = t^2 - 6t + 5$$

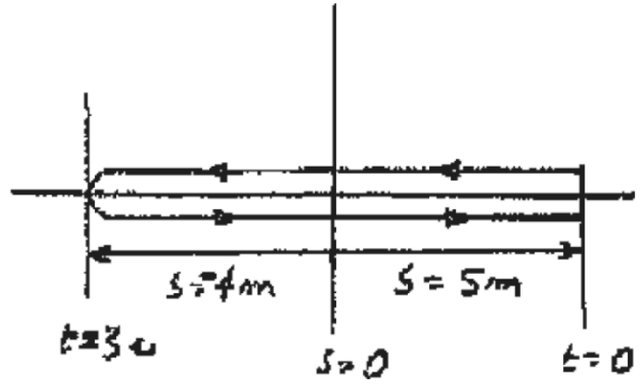
$$v = \frac{ds}{dt} = 2t - 6$$

$$a = \frac{dv}{dt} = 2$$

$v = 0$ when $t = 3$

$$s\big|_{t=0} = 5$$

$$s\big|_{t=3} = -4$$



$$s\big|_{t=6} = 5$$

$$v_{\text{avg}} = \frac{\Delta s}{\Delta t} = \frac{0}{6} = 0$$ **Ans.**

$$(v_{sp})_{\text{avg}} = \frac{s_T}{\Delta t} = \frac{9 + 9}{6} = 3 \text{ m/s}$$ **Ans.**

$$a\big|_{t=6} = 2 \text{ m/s}^2$$ **Ans.**

Ans:

$v_{\text{avg}} = 0$

$(v_{\text{sp}})_{\text{avg}} = 3 \text{ m/s}$

$a\big|_{t=6\text{ s}} = 2 \text{ m/s}^2$